

Nomination for a Chiplet Summit Fellow

Proposed by Prof. David Lariviere and Dr. Paul Borrill

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Introduction

The Chiplet Summit has emerged as the premier venue for dialogue across industry and academia on the rapidly evolving technology of chiplet-based design. To strengthen its role as both a technical and community hub, we propose the establishment of a **Chiplet Summit Fellow Program**. Unlike the “Lifetime Achievement Award” of the Flash Memory Summit, the Fellow Program would recognize **mid-career leaders** with a sustained record of technical contributions, service, and mentorship.

Motivation

- The industry is in the early stages of wide adoption of chiplet integration. This is a time when the most important contributions are being made by rising leaders, not only by established figures nearing retirement.
- A Fellow designation emphasizes both **technical excellence** and **community engagement**, rewarding researchers and engineers at a formative stage of their careers.
- This model aligns more closely with IEEE/ACM traditions and creates continuity of engagement—Fellows can contribute talks, tutorials, and serve on selection committees in subsequent years.

Program Structure

Eligibility Roughly 7–20 years post-Ph.D., with demonstrable contributions to chiplet architectures, packaging, interconnects, or chiplet-aware software and tools.

- Criteria*
1. Technical impact: widely cited publications, prototypes, or open-source artifacts.
 2. Community leadership: ISCA, SIGARCH, HiPChips workshops, or standards activity.
 3. Ecosystem influence: bridging architecture, EDA, packaging, and systems.
 4. Mentorship and inclusion: evidence of training and supporting new entrants to the field.

Process Open nominations, requiring two letters and a concise dossier (bio, five most impactful works, service record). 1–2 Fellows selected annually by a balanced academic/industry committee.

Inaugural Fellow Nominee: Professor Martha A. Kim

We propose **Professor Martha A. Kim** (Columbia University) as the inaugural Chiplet Summit Fellow.

Background

Martha Kim received her Ph.D. in Computer Science from the University of Washington in 2008 under the supervision of Mark Oskin [1]. She is now an Associate Professor of Computer Science at Columbia University, where she chairs the Computer Engineering Program and directs the ARCADE Lab. Her research spans architecture, compilers, accelerators, and systems-level co-design. A current profile can be found on her LinkedIn page [11].

Contributions Relevant to Chiplets

- **Architectural Implications of Brick and Mortar Silicon Manufacturing** (ISCA 2007): A seminal exploration of assembling chips from prefabricated blocks, conceptually aligned with chiplet integration [8].
- **Polymorphic On-Chip Networks** (ISCA 2008): Foundational work on flexible interconnects, anticipating the needs of multi-die and heterogeneous systems [9].
- Long-standing service in the architecture community: ISCA 2014 Tutorial Co-Chair; SIGARCH Executive Committee member.

Why Professor Kim

Professor Kim exemplifies the qualities intended for Chiplet Summit Fellows:

1. Deep technical research with lasting influence on the chiplet concept.
2. Active leadership in the architecture community.
3. A career trajectory that promises further impact at the intersection of academic innovation and industry adoption.

Conclusion

The creation of a **Chiplet Summit Fellow Program** offers a technical, forward-looking complement to marketing-style awards at other conferences. Recognizing **Martha A. Kim** as the inaugural Fellow would immediately set the tone: the Summit is committed to technical excellence, rising leaders, and building a vibrant ecosystem for the next era of semiconductor design.

Appendix: Draft Nomination Letter

To: Organizers of the Chiplet Summit **From:** [Your Name], on behalf of the Technical Community **Subject:** Proposal for a Chiplet Summit Fellow Program and Nomination of Professor Martha A. Kim

Dear Colleagues, It is with great enthusiasm that I propose the establishment of a **Chiplet Summit Fellow Program**. This initiative will serve to recognize outstanding mid-career contributors whose work has advanced the technical and community foundations of chiplet integration. Unlike traditional lifetime achievement awards, this designation will highlight those whose best contributions may still lie ahead. For the inaugural Fellow, I respectfully nominate **Professor Martha A. Kim** of Columbia University. Professor Kim's career embodies the spirit of this recognition:

- Her Ph.D. research at the University of Washington (2008) [1] addressed fine-grained parallelism, including early explorations of "brick and mortar" silicon concepts that prefigured today's chiplet design philosophy.
- She has published influential ISCA papers on modular silicon design and polymorphic interconnects, establishing a strong technical basis for scalable chiplet integration [8, 9].
- At Columbia University, she leads the ARCADE Lab, mentoring the next generation of computer architects and fostering industry-academic collaboration.
- She has demonstrated sustained community leadership as ISCA Tutorial Co-Chair and as a member of the SIGARCH Executive Committee.

Professor Kim's profile on LinkedIn provides further context for her broad experience and leadership in the field [11]. Recognizing her as the first Chiplet Summit Fellow would signal our community's commitment to excellence, innovation, and inclusivity at a formative stage in this technology's history.

References

- [1] Martha A. Kim, *Hardware Support for Efficient, Fine-Grained Parallelism*, Ph.D. Thesis, University of Washington, 2008.
- [2] M. G. Kostoulas, M. Matsa, N. Mendelsohn, E. Perkins, A. Heifets, M. Mercaldi, "XML Screamer: An Integrated Approach to High Performance XML Parsing, Validation and Deserialization," in *Proceedings of the 15th International Conference on World Wide Web (WWW)*, pp. 93–102, 2006.

- [3] M. Mercaldi, S. Swanson, A. Petersen, A. Putnam, A. Schwerin, M. Oskin, S. J. Eggers, "Instruction Scheduling for a Tiled Dataflow Architecture," in Proceedings of the 12th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), pp. 141–150, 2006.
- [4] M. Mercaldi, S. Swanson, A. Petersen, A. Putnam, A. Schwerin, M. Oskin, S. J. Eggers, "Modeling Instruction Placement on a Spatial Architecture," in Proceedings of the 18th Annual ACM Symposium on Parallelism in Algorithms and Architectures (SPAA), pp. 158–169, 2006.
- [5] A. Petersen, A. Putnam, M. Mercaldi, A. Schwerin, S. Swanson, S. J. Eggers, M. Oskin, "Reducing Control Overhead in Dataflow Architectures," in Proceedings of the 15th International Conference on Parallel Architectures and Compilation Techniques (PACT), pp. 182–191, 2006.
- [6] S. Swanson, A. Putnam, M. Mercaldi, K. Michelson, A. Petersen, A. Schwerin, M. Oskin, S. J. Eggers, "Area-Performance Trade-offs in Tiled Dataflow Architectures," in Proceedings of the 33rd International Symposium on Computer Architecture (ISCA), pp. 314–326, 2006.
- [7] S. Swanson, A. Schwerin, M. Mercaldi, A. Petersen, A. Putnam, K. Michelson, M. Oskin, S. J. Eggers, "The WaveScalar Architecture," ACM Transactions on Computer Systems (TOCS), vol. 25, no. 2, pp. 4:1–4:54, 2007.
- [8] M. Mercaldi Kim, M. Mehrara, M. Oskin, T. Austin, "Architectural Implications of Brick and Mortar Silicon Manufacturing," in Proceedings of the 34th International Symposium on Computer Architecture (ISCA), pp. 244–253, 2007.
- [9] M. Mercaldi Kim, J. D. Davis, M. Oskin, T. Austin, "Polymorphic On-Chip Networks," in Proceedings of the 35th International Symposium on Computer Architecture (ISCA), pp. 101–112, 2008.
- [10] M. A. Kim, S. A. Edwards, "Computation vs. Memory Systems: Pinning Down Accelerator Bottlenecks," in Proceedings of the 3rd Workshop on Architectural and Microarchitectural Support for Binary Translation (AMAS-BT), 2010.
- [11] Martha A. Kim, LinkedIn Profile. <https://www.linkedin.com/in/marthakim>

- [12] M. Kambadur, K. Tang, M. A. Kim, "Harmony: Collection and Analysis of Parallel Block Vectors," in Proceedings of the 39th International Symposium on Computer Architecture (ISCA), pp. 452–463, 2012.
- [13] M. Kambadur, T. Moseley, R. Hank, M. A. Kim, "Measuring Interference Between Live Datacenter Applications," in Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2012.
- [14] L. Wu, R. J. Barker, M. A. Kim, K. A. Ross, "Navigating Big Data with High-Throughput, Energy-Efficient Data Partitioning," in Proceedings of the 40th International Symposium on Computer Architecture (ISCA), pp. 480–491, 2013.
- [15] L. Wu, A. Lottarini, T. K. Paine, M. A. Kim, K. A. Ross, "Q100: The Architecture and Design of a Database Processing Unit," in Proceedings of the 19th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), pp. 255–268, 2014.
- [16] M. Kambadur, M. A. Kim, "An Experimental Survey of Energy Management Across the Stack," in Proceedings of the 2014 ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA), pp. 329–344, 2014.
- [17] L. Wu, A. Lottarini, T. K. Paine, M. A. Kim, K. A. Ross, "The Q100 Database Processing Unit," IEEE Micro, vol. 35, no. 3, pp. 34–46, 2015.
- [18] M. Kambadur, S. Hong, J. Cabral, H. Patil, C.-K. Luk, S. Krishnamoorthy, M. A. Kim, "Fast Computational GPU Design with GT-Pin," in Proceedings of the International Symposium on Workload Characterization (IISWC), pp. 76–86, 2015.
- [19] R. Townsend, C. Wu, M. A. Kim, S. A. Edwards, "From Functional Programs to Pipelined Dataflow Circuits," in Proceedings of the 26th International Conference on Compiler Construction (CC), pp. 76–86, 2017.
- [20] R. Townsend, M. A. Kim, S. A. Edwards, "Pipelining a Triggered Processing Element," in Proceedings of the 50th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO), pp. 357–368, 2017.
- [21] A. Lottarini, A. Ramirez, J. Coburn, M. A. Kim, P. Ranganathan, D. Stodolsky, M. Wachsler, "vbench: Benchmarking Video

- Transcoding in the Cloud," in Proceedings of the 23rd International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), pp. 797–809, 2018.
- [22] M. A. Kim, Y. S. Shao, "Hardware Acceleration," *IEEE Micro*, vol. 38, no. 6, pp. 6–7, 2018.
 - [23] J. P. Cerqueira, T. J. Repetti, Y. Pu, S. Priyadarshi, M. A. Kim, M. Seok, "Catena: A 0.5-V Sub-0.4-mW 16-Core Spatial Array Accelerator for Mobile and Embedded Computing," in Proceedings of the Symposium on VLSI Circuits, pp. 54–55, 2019.
 - [24] A. Lottarini, J. P. Cerqueira, T. J. Repetti, S. A. Edwards, K. A. Ross, M. Seok, M. A. Kim, "Master of None Acceleration: A Comparison of Accelerator Architectures for Analytical Query Processing," in Proceedings of the 46th International Symposium on Computer Architecture (ISCA), pp. 762–773, 2019.
 - [25] S. A. Edwards, R. Townsend, M. Barker, M. A. Kim, "Compositional Dataflow Circuits," *ACM Transactions on Embedded Computing Systems (TECS)*, vol. 18, no. 1, pp. 5:1–5:27, 2019.
 - [26] T. Guan, P. Liu, X. Zeng, M. A. Kim, M. Seok, "Recursive Binary Neural Network Training Model for Efficient Usage of On-Chip Memory," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 66, no. 7, pp. 2593–2605, 2019.
 - [27] J. P. Cerqueira, T. J. Repetti, Y. Pu, S. Priyadarshi, M. A. Kim, M. Seok, "Catena: A Near-Threshold, Sub-0.4-mW, 16-Core Programmable Spatial Array Accelerator for the Ultralow-Power Mobile and Embedded Internet of Things," *IEEE Journal of Solid-State Circuits*, vol. 55, no. 8, pp. 2270–2284, 2020.
 - [28] M. Barker, S. A. Edwards, M. A. Kim, "Synthesized Garbage Collection for FPGA Accelerators," in Proceedings of the 30th ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA), p. 53, 2022.
 - [29] M. Barker, S. A. Edwards, M. A. Kim, "Synthesized In-BRAM Garbage Collection for Accelerators with Immutable Memory," in Proceedings of the 32nd International Conference on Field-Programmable Logic and Applications (FPL), pp. 47–53, 2022.
 - [30] C. Espenshade, R. Peng, E. Hong, M. Calman, Y. Zhu, P. Parida, E. K. Lee, M. A. Kim, "Characterizing Training Performance and Energy for Foundation Models and Image Classifiers on Multi-Instance GPUs," *arXiv preprint arXiv:2312.08955*, 2023.

- [31] P. K. Chundi, Y. Zhou, M. A. Kim, M. Seok, "Hotspot Monitoring and Temperature Estimation with Miniature On-Chip Temperature Sensors," in Proceedings of the Design, Automation & Test in Europe Conference & Exhibition (DATE), 2024.
- [32] L. Eeckhout, G. Smaragdakis, K. Liang, A. Sampson, M. A. Kim, C. J. Rossbach, eds., "Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 1," ASPLOS 2025, Rotterdam, The Netherlands, ACM, ISBN 979-8-4007-0698-1, 2025.
- [33] L. Eeckhout, G. Smaragdakis, K. Liang, A. Sampson, M. A. Kim, C. J. Rossbach, eds., "Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2," ASPLOS 2025, Rotterdam, The Netherlands, ACM, ISBN 979-8-4007-1079-7, 2025.